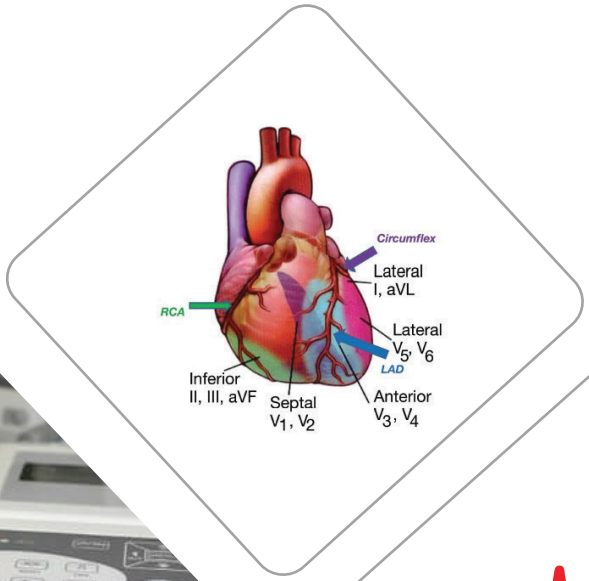
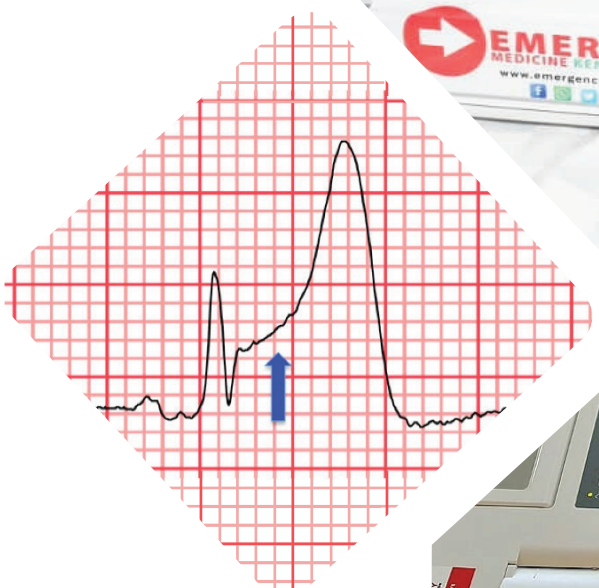
STEMI ECG INTERPRETATION











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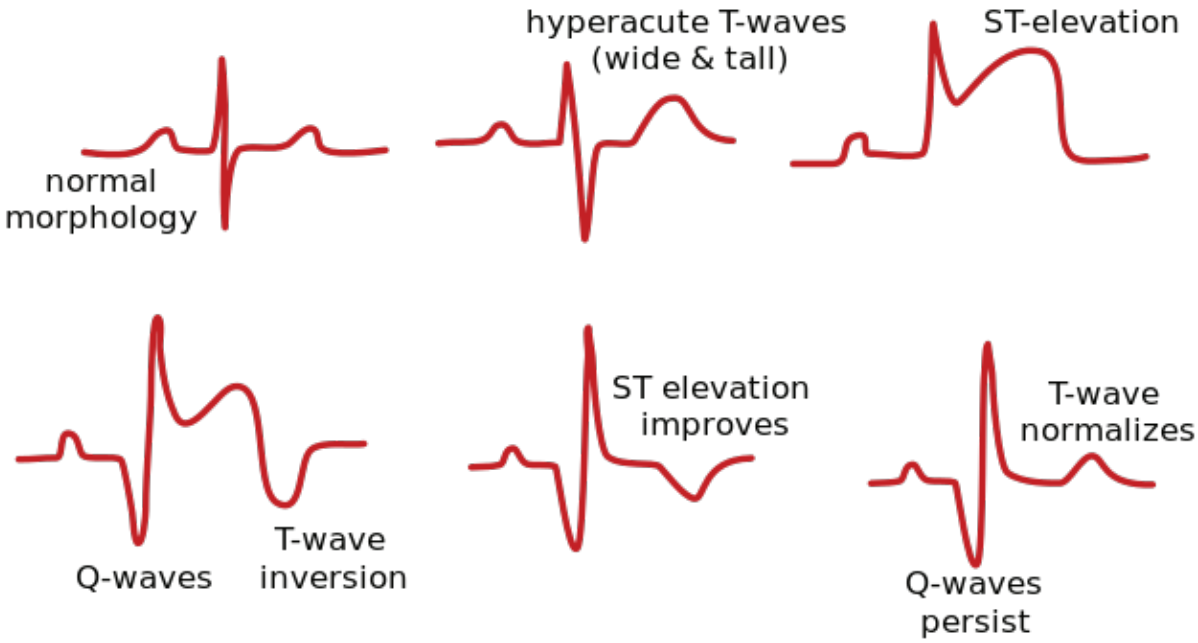
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Introduction

In the early stages of acute myocardial infarction (MI), the electrocardiogram may be normal or near normal; **< ½** of patients with acute myocardial infarction have clear diagnostic changes on their **first trace**.

About 10% of patients with a proved acute myocardial infarction (based on clinical history and enzymatic markers) **fail to develop ST segment elevation or depression**. In most cases, however, serial electrocardiograms show evolving changes that tend to follow well recognised patterns.

Sequence of changes seen during evolution of myocardial infarction

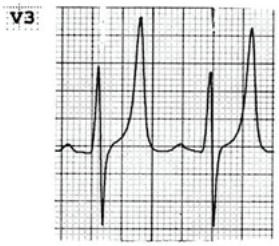


Hyperacute T waves

The **earliest signs** of acute myocardial infarction are subtle and include **increased T wave amplitude** over the affected area.

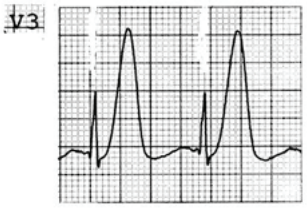
T waves become more prominent, **symmetrical,** and pointed ("hyperacute"). Hyperacute T waves are most evident in the **anterior chest leads** and are more readily visible when an old electrocardiogram is available for comparison. These changes in T waves are usually present for **only 5 to 30 minutes** after the onset of the infarction and are followed by ST segment changes.

Hyperkalemia



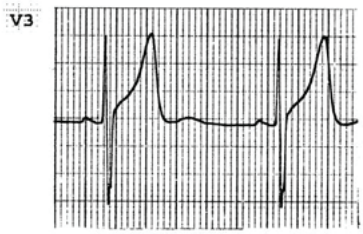
Symmetric, narrow-based, pointed, tenting

Hyperacute Ischemia



Symmetric, broad-based, not tented, not pointed
QT interval tends to be long (not in this example)

Normal Variant



Asymmetric and not narrow

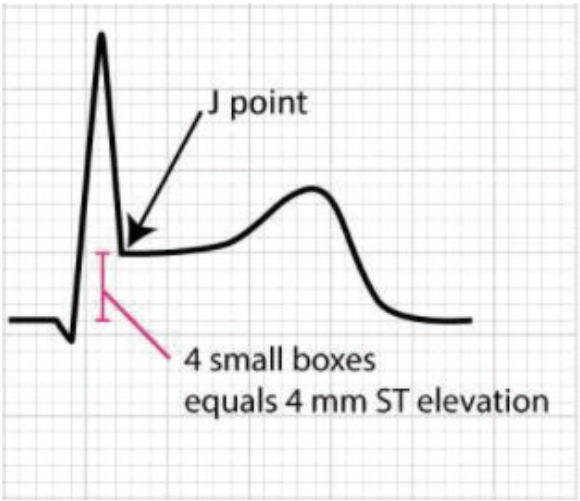
ECG diagnostic criteria of AMI

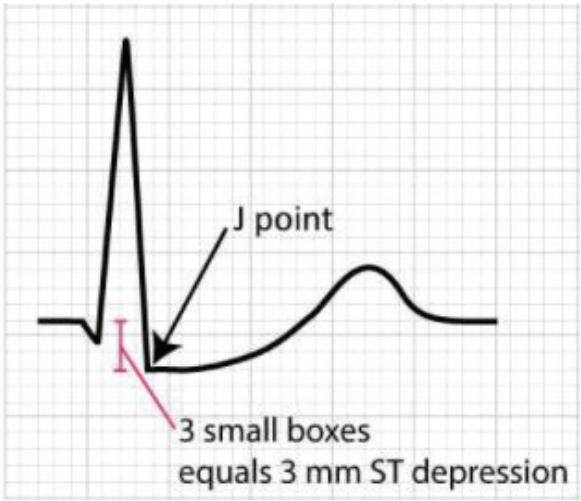
ST elevation MI (STEMI) is a clinical syndrome defined by characteristic symptoms of myocardial ischemia in association with persistent electrocardiographic (ECG) ST elevation and subsequent release of biomarkers of myocardial necrosis. Diagnostic ST elevation in the absence of left ventricular (LV) hypertrophy or left bundle-branch block (LBBB) is defined as ST elevation **at the J point** in at least **2 contiguous leads** of;

- **≥ 2 mm (0.2 mV) in men (2.5 mm if < 40yrs) or ≥ 1.5 mm (0.15 mV) in women in leads V2–V3**

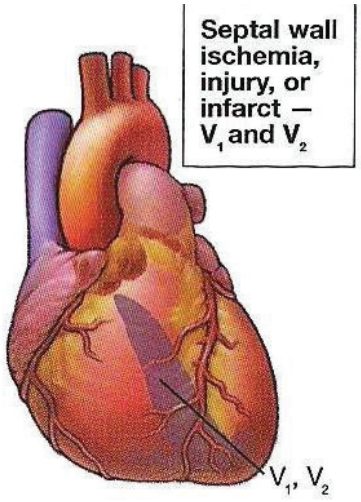
and/or

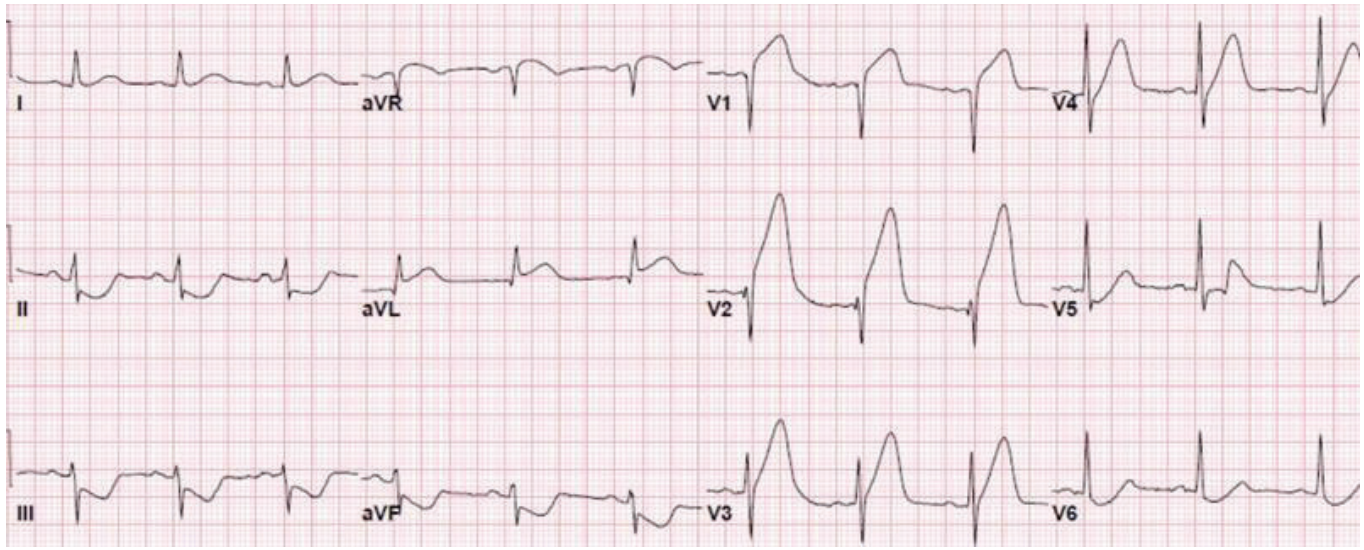
- **≥ 1 mm (0.1mV) in other contiguous chest leads or the limb leads**





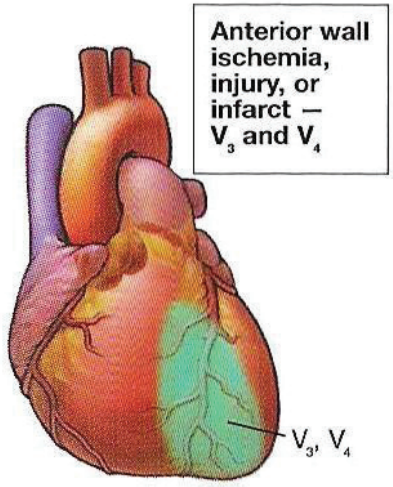
Septal STEMI

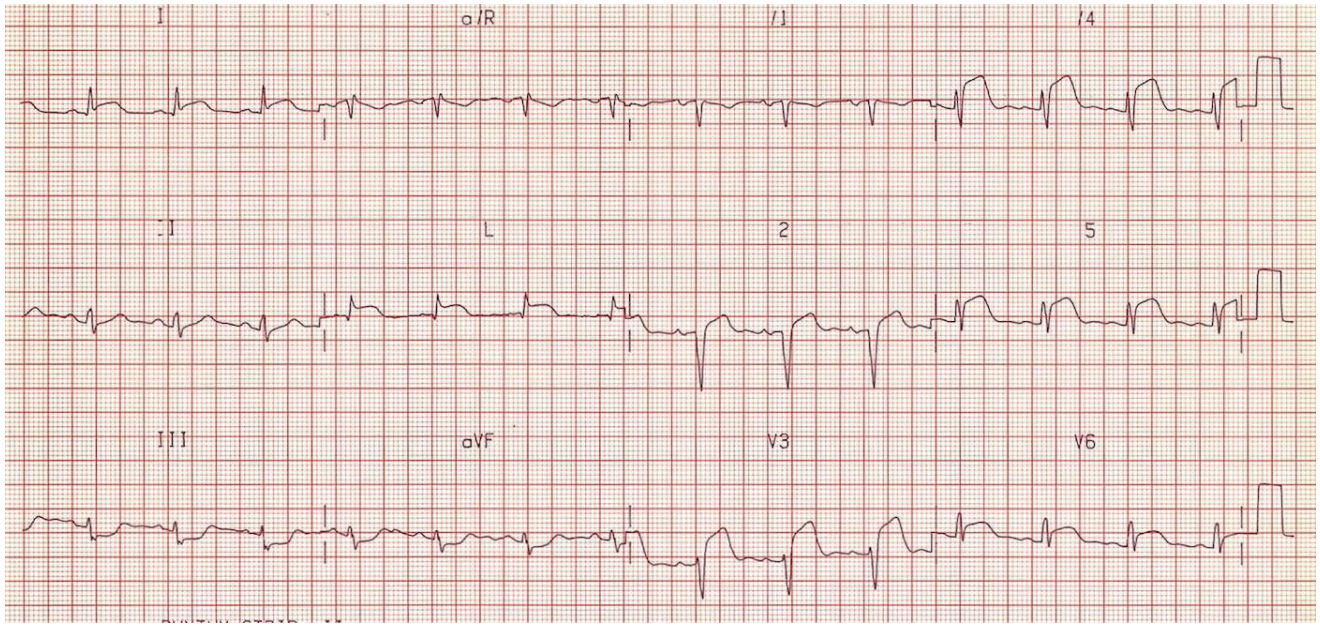




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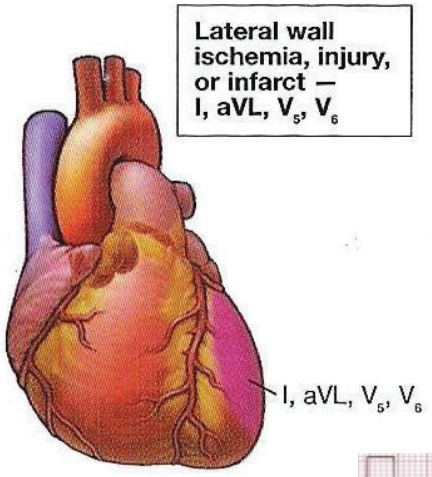
Anterior STEMI

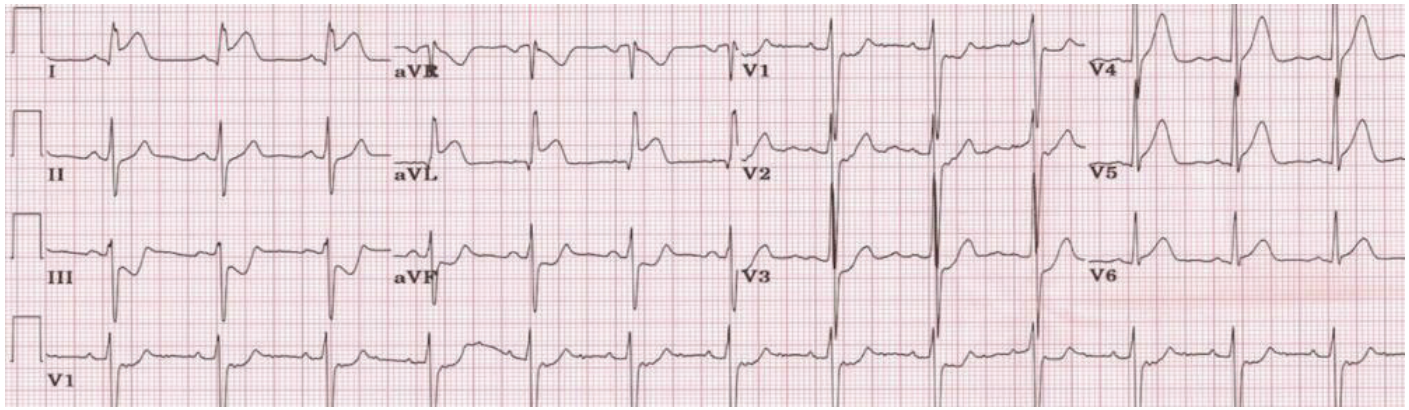




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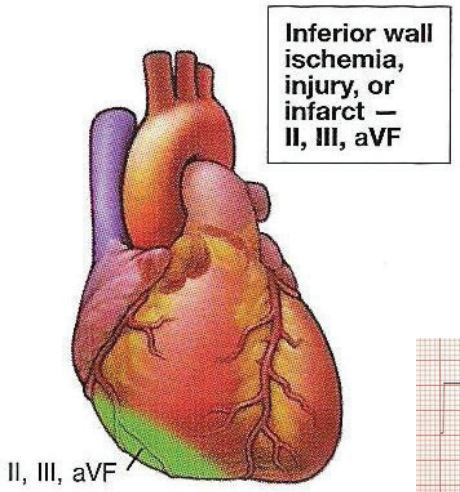
Lateral STEMI

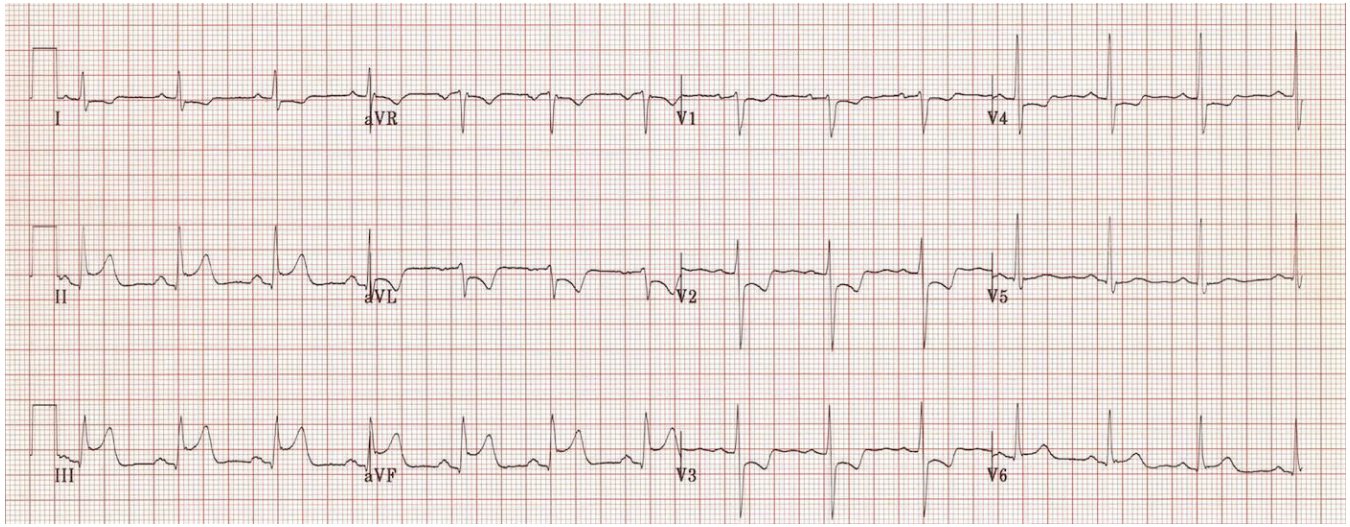




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Inferior STEMI





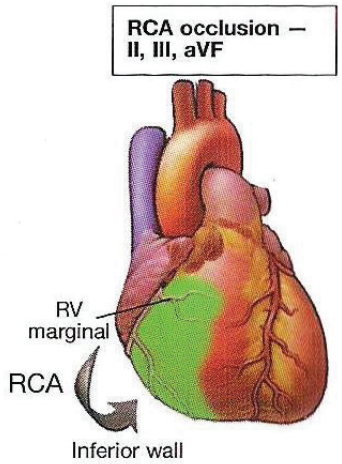
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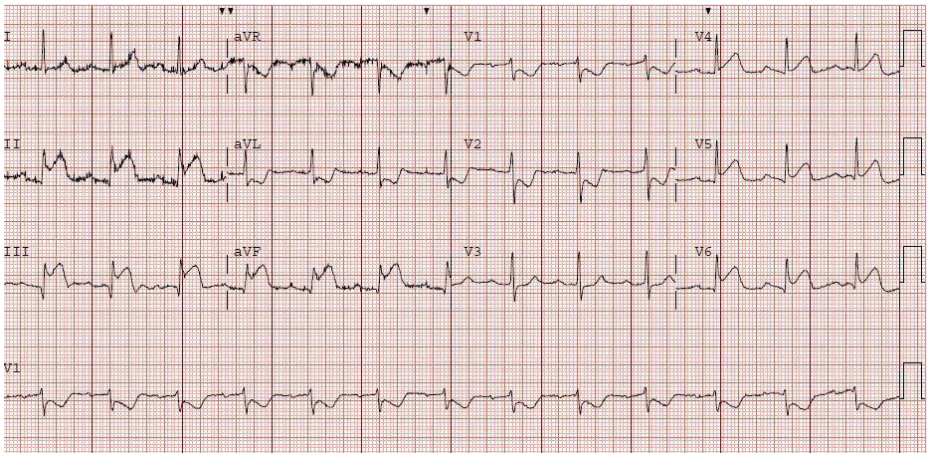
Right-sided STEMI

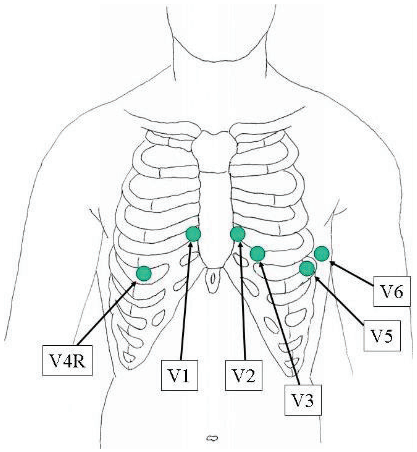
Right ventricular infarction usually results from occlusion of the **right coronary artery proximal to the right ventricular marginal branches**, hence its association with **50% of inferior infarctions.** Less commonly, right ventricular infarction is associated with occlusion of the **distal circumflex artery**, and if this vessel is dominant there may be an associated **inferolateral wall** infarction.

On the standard 12 lead electrocardiogram right ventricular infarction is indicated by;

- **STE** in lead **V_1** with simultaneous **ST depression** in **V_2**, is almost always associated with RV involvement—recall that lead V_1 essentially serves as a right-sided lead.
- Signs of **inferior infarction** - the magnitude of STE in lead III relative to the other inferior leads, that is, when the **STE** in lead **III** is significantly more prominent than in lead **II** (e.g., **> 2-3 mm more prominent**), an **RV MI is highly likely**.





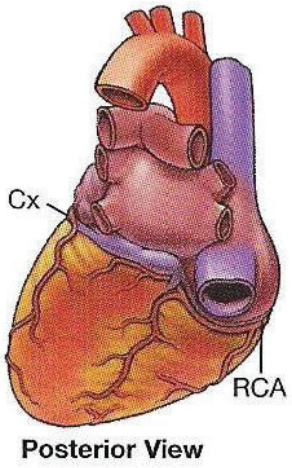


The clinician should suspect **RV infarction** in patients with **inferior wall infarction**, **hypotension**, and **clear lung fields**. In patients with inferior wall infarction, obtain an ECG with right-sided leads. ST-segment elevation (**>1 mm**) in lead **V_4R** is sensitive (**sensitivity, 88%; specificity, 78%**; **diagnostic accuracy, 83%**) for **RV infarction** and is a **strong predictor of increased in-hospital complications and mortality.**

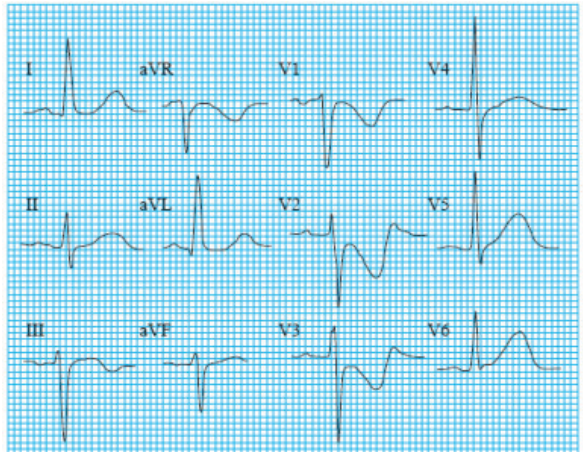
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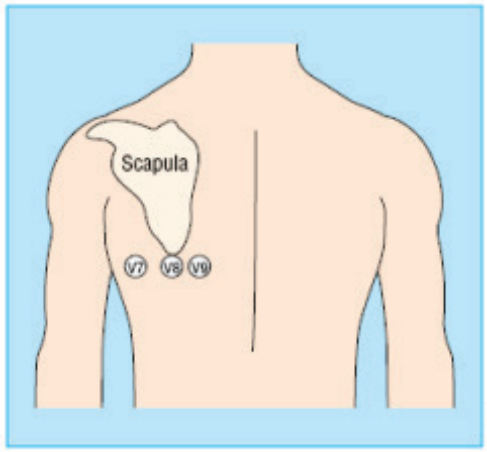
Posterior STEMI

Posterior myocardial infarction refers to infarction of the **posterobasal wall of the left ventricle.** Approximately **4%** of acute MI patients have STE isolated to the posterior leads. Isolated posterior STEMI does qualify for **emergent reperfusion therapy**; therefore, obtain posterior leads on patients when the initial standard ECG is non-diagnostic and the clinical suspicion for infarction is high.



Posterior View





Position of V7, V8, and V9 on posterior chest wall

The changes of posterior myocardial infarction are seen indirectly in the anterior precordial leads. **Leads V_1 to V_3** face the **endocardial surface** of the **posterior wall of the left ventricle**. As these leads record from the opposite side of the heart instead of directly over the infarct, the changes of posterior infarction are reversed in these leads.

- The **R waves increase in size**, becoming broader and dominant, and are associated with **ST depression** and **upright T waves.** This contrasts with the Q waves, ST segment elevation, and T wave inversion seen in acute anterior myocardial infarction.

Ischaemia of the **anterior wall of the left ventricle** also produces **ST segment depression in leads V_1 to V_3,** and this must be differentiated from posterior myocardial infarction. The use of posterior leads **V_7 to V_9** will show **ST segment elevation** in patients with **posterior infarction**.

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Chest Pain (Acute Coronary Syndrome) Algorithm

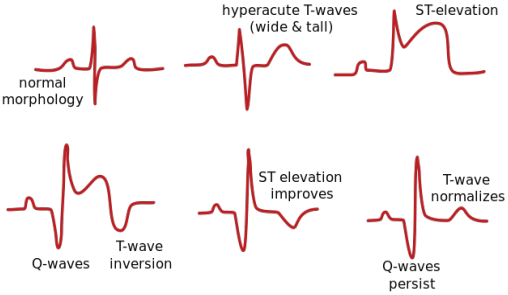
This clinical pathway is intended to supplement, rather than substitute for, professional judgment and may be changed depending upon a patient's individual needs. Failure to comply with this pathway does not represent a breach of the standard of care.

Chest Discomfort Suggestive of Ischemia
(includes **anginal equivalents** (atypical symptoms) like exertional pain in the ear, jaw, neck, shoulder, arm, back, or epigastric area; exertional dyspnoea; nausea and vomiting; diaphoresis; and fatigue.

- Monitor, support ABCs in the **Resuscitation Room (ER)**. Be prepared to provide CPR, Defibrillation and **?Thrombolysis/Fibrinolysis**
- **Obtain/review 12-lead ECG within 10 minutes of arrival to ED**
  - Do a **V4R** if ST elevation in lead V1 with simultaneous ST depression in V2 -? **Right sided STEMI**
  - Do **V7 - V9** if ST depressions ≥ 1 mm with upright T-waves in ≥ 2 contiguous anterior precordial leads (V1 to V3) -? **Posterior STEMI**
  - If there is **ST elevation in aVR ≥ 1mm** and **aVR ≥ V1** with widespread horizontal ST depression, most prominent in leads I, II and V4-6 – **consult an Interventional Cardiologist** immediately for PCI (**Left main coronary artery occlusion/Proximal LAD lesion/Severe sub endocardial ischaemia, nonlocalized**)
  - Sinus Tachycardia, T wave inversion in III & V1, V3 or (S1, Q3, T3) pattern - Consider **Pulmonary Embolism**
- Check vital signs (BP, PR, RR, SPO_2, T°C, **RBS**)
- Start Oxygen **IF** SPO_2 < 90% or if patient is dyspnoeic. Maintain SPO_2 **≥** 90%
- Perform brief, targeted history, physical exam – **Indicate time of symptoms onset**

- **Consider other life-threatening causes of chest pain** (**pulmonary embolus, cardiac tamponade, aortic dissection, tension pneumothorax, oesophageal rupture**)
- **Review initial 12-lead ECG**

Sequence of ECG changes seen during evolution of myocardial infarction - In the early stages of acute myocardial infarction the electrocardiogram may be normal or near normal; **< ½ of patients with acute myocardial infarction have clear diagnostic changes on their first trace**. **About 10%** of patients with a proved acute myocardial infarction (on the basis of clinical history and enzymatic markers) fail to develop ST segment elevation or depression. In most cases, however, serial electrocardiograms show evolving changes that tend to follow well recognised patterns.



ST Elevation	MI Description	Coronaries affected
V2 – V5	Anterior	LAD
V1 – V2	Septal	Septal LAD
II, III, aVF	Inferior	RCx (20%) or RCA (80%)
V1 – V4	**Anteroseptal**	
V3 – V6, I, aVL	**Anterolateral**	
I, aVL, V5, V6	Lateral	LCx
V7, V8, V9	Posterior	RCx
V1, V4R	RV	RCA

***** LAD, Left Anterior Descending; RCx, Right Circumflex; RCA, Right Coronary Artery; LCx, Left Circumflex; V4R, Right sided V4.

- **Sgarbossa's Criteria** for patients with **Left Bundle Branch Blocks (LBBB)** available in **MDCalc**

ST elevation
ST-Elevation MI (STEMI)
ST elevation at the J point in at least two contiguous leads of ≥ 1 mm

→ **See STEMI Algorithm**

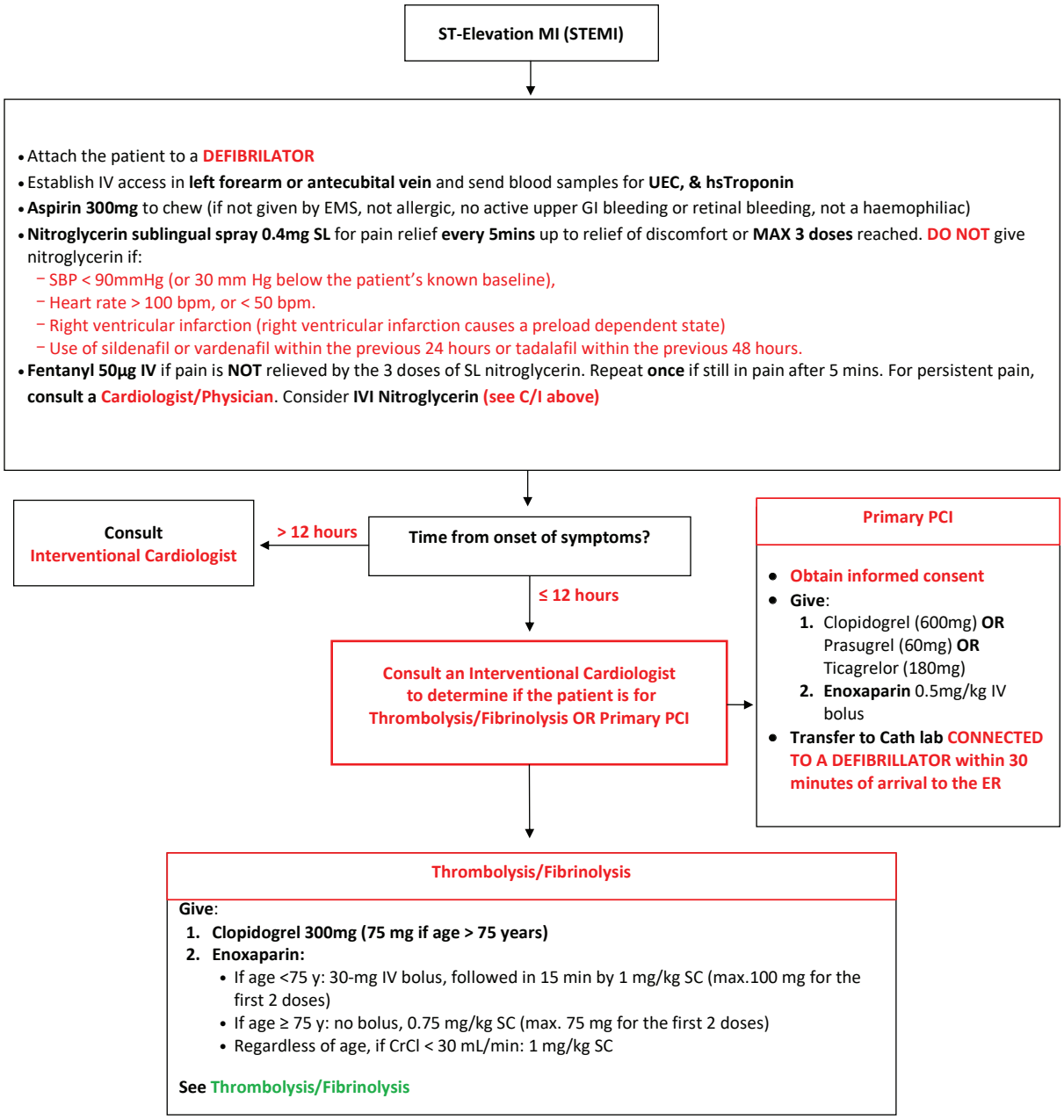
ST depression > 0.5mm or dynamic T-wave inversion ≥ 2mm; strongly suspicious for ischemia
High-Risk Unstable Angina/Non-ST-Elevation MI (UA/NSTEMI)

Normal or Non-diagnostic changes in ST segment or T wave
Intermediate/Low Risk UA

→ **See NSTEMI/UA Algorithm**

STEMI Algorithm

This clinical pathway is intended to supplement, rather than substitute for, professional judgment and may be changed depending upon a patient's individual needs. Failure to comply with this pathway does not represent a breach of the standard of care.



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Thrombolysis/Fibrinolysis

This clinical pathway is intended to supplement, rather than substitute for, professional judgment and may be changed depending upon a patient's individual needs. Failure to comply with this pathway does not represent a breach of the standard of care.

Review/Complete Fibrinolysis

Absolute Contraindications

- □ Any prior intracranial haemorrhage
- □ Known structural cerebral vascular lesson (e.g., AVM)
- □ Known malignant intracranial neoplasm (primary or metastatic)
- □ Ischaemic stroke within 6 months EXCEPT acute ischaemic stroke within 4.5 hours
- □ Suspected aortic dissection
- □ Active bleeding or bleeding diathesis (excluding menses)
- □ Significant closed-head trauma or facial trauma within 1 month
- □ Intracranial or intraspinal surgery within 1 month
- □ Severe uncontrolled hypertension (unresponsive to emergency therapy)
- □ For streptokinase, prior treatment with streptokinase within the previous 6 months
- □ Non-compressible vascular punctures in the past 24 hrs

Relative Contraindications

- □ Severe hypertension on presentation (SBP > 180 mmHg or DBP > 110 mmHg) - Lower BP first before fibrinolysis
- □ History of TIA in the preceding 6 months
- □ Traumatic or prolonged (>10 minutes) CPR
- □ Major surgery (< 3 weeks)
- □ Recent (within 2 to 4 weeks) internal bleeding
- □ Pregnancy or within 1 week post-partum
- □ Active peptic ulcer
- □ Oral anticoagulant therapy

No contraindications for Thrombolysis/Fibrinolysis

- **Obtain informed consent for fibrinolysis/thrombolysis**
- Ensure the patient is **connected to a defibrillator** (ECG, SPO_2, BP) and repeat baseline vitals. **Administer fibrinolysis/thrombolysis within 10 mins of STEMI diagnosis**

Fibrinolytic Agent	Dose	Fibrin Specificity*	Antigenic	Patency Rate (90-min TIMI 2 or 3 flow)
Fibrin-specific:				
Tenecteplase (TNK-tPA) ***Half dose in patients ≥75 yrs)**	To reconstitute, mix the 50-mg vial in 10 mL sterile water (5 mg/mL). Give IV bolus based on weight as below: < 60 kg - 30 mg (6 mL) 60 to 69 kg - 35 mg (7 mL) 70 to 79 kg - 40 mg (8 mL) 80 to 89 kg - 45 mg (9 mL) ≥ 90 kg - 50 mg (10 mL)	+ + + +	No	85%
Reteplase (rPA)	10 U+10-U IV boluses given 30 min apart	+ +	No	84%
Alteplase (tPA)	Bolus 15 mg IV, then give an infusion of 0.75 mg/kg for 30 min (maximum 50 mg), then 0.5 mg/kg (maximum 35 mg) over the next 60 min; total dose not to exceed 100 mg.	+ +	No	73% to 84%
Non–fibrin-specific:				
Streptokinase	Set up a second IV line for Streptokinase. The adult dose of streptokinase for STEMI is 1.5 Million U in 50 mL of 5% dextrose in water (D5W) given IV over 30-60 minutes. Allergic reactions force the termination of many infusions before a therapeutic dose can be administered. Run Ringer's Lactate/Hartmann's Solution TKVO in the other line	No	Yes§	60% to 68%

*Strength of fibrin specificity; "+ + + +" is more strong, "+ +" is less strong.
§Streptokinase is highly antigenic and absolutely contraindicated within 6 mo of previous exposure because of the potential for serious allergic reaction.

IV indicates intravenous; rPA, reteplase plasminogen activator; TIMI, Thrombolysis In Myocardial Infarction; TNK-tPA, tenecteplase tissue-type plasminogen activator; and tPA, tissue-type plasminogen activator.

- Monitor vital signs (BP, PR, RR, SPO_2) **every 15 minutes** during the infusions
- Continue monitoring the patient for **30mins after the end of the infusions**
- **Transfer patient to CCU/ICU CONNECTED TO A DEFIBRILLATOR**

NSTEMI/Unstable Angina Algorithm

This clinical pathway is intended to supplement, rather than substitute for, professional judgment and may be changed depending upon a patient's individual needs. Failure to comply with this pathway does not represent a breach of the standard of care.

ST depression > 0.5mm or dynamic T-wave inversion ≥ 2mm; strongly suspicious for ischemia
High-Risk Unstable Angina/Non-ST-Elevation MI (UA/NSTEMI)

Normal or Non-diagnostic changes in ST segment or T wave
Intermediate/Low Risk UA

- Establish IV access and send blood samples for **UEC, & hsTroponin** **(obtain hsTroponin at least 4 hours after symptom onset, not before)**
- **Aspirin 300mg** to chew (if not given by EMS, not allergic, no active upper GI bleeding or retinal bleeding, not a haemophiliac)
- **Nitroglycerin sublingual spray 0.4mg SL** for pain relief **every 5mins** up to relief of discomfort or **MAX 3 doses** reached. **DO NOT** give nitroglycerin if:
  - SBP < 90mmHg (or 30 mm Hg below the patient's known baseline),
  - Heart rate > 100 bpm, or < 50 bpm.
  - Right ventricular infarction (right ventricular infarction causes a preload dependent state)
  - Use of sildenafil or vardenafil within the previous 24 hours or tadalafil within the previous 48 hours.
- **Fentanyl 50µg IV** if pain is **NOT** relieved by the 3 doses of SL nitroglycerin. Repeat **once** if still in pain after 5 mins. For persistent pain, **consult a Cardiologist/Physician**. Consider **IVI nitroglycerin** **(see C/I above)**
- **Consider CXR**

* available in MDCalc

The HEART Score for Chest Pain Patients in the ED		
History	• Highly Suspicious • Moderately Suspicious • Slightly or Non-Suspicious	• 2 points • 1 point • 0 points
ECG	• Significant ST-Depression • Nonspecific Repolarization • Normal	• 2 points • 1 point • 0 points
Age	• ≥ 65 years • > 45 - < 65 years • ≤ 45 years	• 2 points • 1 point • 0 points
Risk Factors	• ≥ 3 Risk Factors or History of CAD • 1 or 2 Risk Factors • No Risk Factors	• 2 points • 1 point • 0 points
Troponin	• ≥ 3 x Normal Limit • > 1 - < 3 x Normal Limit • ≤ Normal Limit	• 2 points • 1 point • 0 points
Risk Factors: DM, current or recent (<one month) smoker, HTN, HLP, family history of CAD, & obesity		
Score 0 – 3: 1.7% MACE over next 6 weeks; **Score 4 – 6:** 16.6% MACE over next 6 weeks; **Score 7 – 10:** 50.1% MACE over next 6 weeks		

Backus BE et al. A Prospective Validation of the HEART Score for Chest Pain Patients at the Emergency Department. International Journal of Cardiology. 2013. 168; 2153 – 2158. PMID: 23465250

Repeat 12 lead ECG **in 30 mins**/Continuous ST-segment monitoring → **STEMI** **See STEMI Algorithm**

hsTroponin > lab cutoff OR HEART score > 3 points OR ST depression OR Dynamic ECG changes

Yes → **Consult a Cardiologist/Physician**

No → If hsTroponin done < **4 hours** from symptom onset, then **repeat the hsTroponin at least 4 hours after symptom onset.**

If hsTroponin ≥ lab cutoff → **Consult a Cardiologist/Physician**

hsTroponin < lab cutoff

If no evidence of ischaemia or infarction by testing, can discharge ± cardiology follow-up

Criteria for Discharge (must meet ALL the criteria)
- **Two ECGs** which are **Normal** and **No ST Depression** and **No Dynamic changes**
- **hsTroponin < lab cutoff** at **≥ 4 hours** from onset of symptoms **OR** 3 hours from admission to ER

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Did You Know?

The **Constitution of Kenya (2010)** and the **Health Act (2017)** guarantees you

the right to emergency medical treatment

All **public and private health facilities** have a legal duty to provide you with **emergency medical treatment**

Any **Health Institution** that **fails** to provide **emergency medical treatment** despite having the capacity to do so, could face **conviction** and **fines up to Ksh. 3 million**

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